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Protocol for Fit4Study: A non-randomised controlled trial of a physical activity intervention for university students' mental health and wellbeing

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A R T I C L E I N F O

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S T R U C T U R E D A B S T R A C T

Purpose of Research: University students experience higher rates of mental ill health compared to other age groups, with physical activity emerging as a potential preventative strategy. However, few physical activity interventions for university students are designed using behaviour change theories, limiting their potential effectiveness. The purpose of this paper is to outline the protocol for a non-randomised controlled trial evaluating Fit4Study, a physical activity intervention aimed at improving mental health outcomes in undergraduate students, informed by the Capability, Opportunity and Motivation model of behaviour change (COM-B).
Procedure: A 2×2 (condition: Fit4Study, Control Group; time: pre, post) repeated measures ANOVA will be used to analyse the effectiveness of Fit4Study in improving mental health outcomes compared to a no-intervention control group across two time points. Primary outcomes include symptoms of anxiety and depression, psychological wellbeing, loneliness, and use and knowledge of physical activity to support mental health and wellbeing. Secondary outcomes include physical activity and sedentary behaviour. Feasibility will be assessed quantitatively via reach and retention and qualitative data will be collected to analyse participants' acceptability of the content and delivery of Fit4Study.
Main Results: No main results are reported as it is a protocol paper.
New Findings: If the results are significant, Fit4Study has the potential to serve as an easily replicable physical activity intervention aimed at improving university students' mental health and wellbeing. It will also be among the first mental health-focused physical activity interventions to incorporate behaviour change theories in the design and implementation phases.

1. Introduction

University-aged students, typically aged between 18 and 24 years, tend to be more susceptible to mental ill health when compared to individuals in other age groups (Australian Bureau of Statistics, 2022). In a recent observational study of university students, most participants screened positive for mental health problems, and this rate rose in both domestic students (from 66 % to 76 %) and international students (from 46 % to 67 %) during the first wave of the COVID-19 pandemic (Dingle et al., 2022). While psychological interventions are effective, not all who experience mental ill health require treatment. What is required are evidence-based early intervention strategies to prevent and manage the onset of milder symptoms of mental illness (Dingle et al., 2022). There is sufficient evidence to support physical activity as one suitable preventative strategy for promoting university students' mental health and wellbeing (Hrafnkelsdottir et al., 2018; Reyes-Molina et al., 2022; Román-Mata et al., 2020).

University students who are sufficiently physically active have been found to experience fewer symptoms of depression (Hrafnkelsdottir et al., 2018), lower psychological stress (Román-Mata et al., 2020) and increased psychological wellbeing (Reyes-Molina et al., 2022). However, the overall effectiveness of existing physical activity interventions on

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mental health is somewhat varied (Huang et al., 2024). A recent systematic review and meta-analysis by Huang et al. (2024) highlighted the positive but varied effects of physical activity interventions in reducing symptoms of anxiety, depression, and stress among undergraduate students because of high heterogeneity across studies. The review found that these physical activity interventions varied in terms of the type, duration, and intensity of physical activity, which could be explained by the absence of behaviour change theories informing these interventions (Huang et al., 2024. Notably, only five out of the 59 included studies reported behaviour change theories in their design and delivery. Given the mental health benefits of physical activity, it is crucial to develop and evaluate interventions specifically tailored to undergraduate students and how they use physical activity for their mental health, grounded in well-established behaviour change theories.

The Capability, Opportunity, and Motivation model of behaviour change (COM-B) is a behaviour change theory proposed by Michie et al. (2009). It states that behaviours can be changed by building on physical and psychological capability, which includes knowledge, skills, and mental capacity, providing social and environmental opportunities such as social support and access to physical environments where the behaviour is possible, and improving automatic and reflective motivation by providing skills such as planning and how to change habits (Michie et al., 2009). By extension, intervention functions such as persuasion, enablement, and education are general categories used to change behaviour. In applying the COM-B model of behaviour change, Michie et al. (2013) provide a taxonomy of 93 behaviour change techniques (BCTs), which are observable, reproducible, and basic components of a behaviour change intervention. The three components of COM-B, intervention functions, and behaviour change techniques are separate but closely linked; beginning at the simplest level of capability, opportunity, and motivation, and progresses in specificity and complexity from intervention functions to behaviour change techniques (Michie et al., 2013). For example, providing a demonstration of a strength-based exercise increases physical capacity, and along with feedback on performance, is a form of education and training. This approach is then best described as instruction on how to perform behaviour (BCTT 4.1) and feedback on performing behaviour (BCTT 2.2) according to the BCT taxonomy (Michie et al., 2013).

This study will evaluate a 4-session group physical activity intervention called Fit4Study (see details in the methods section). To address the lack of behavior change theory-informed interventions, it was informed by the COM-B model, with intervention components mapped on to the BCTs provided by Michie et al. (2013) to provide a replicable and theory-based physical activity intervention for university students' mental health and wellbeing. Fit4Study was designed as part of a package of mental health-focused early interventions collectively titled 'Sharper Minds'. The Sharper Minds package aims to increase university students' engagement in self-care behaviours such as better sleep, study strategies, nutrition, social support, mood regulation, and physical activity, which have been shown to relate to improved mental health in university students (Dingle et al., 2024). Sharper Minds includes a website with videos and tip sheets, a weekly self-monitoring e-tool tracking these six self-care behaviours, and a choice of short courses to promote the self-care behaviours (Dingle et al., 2021); Fit4Study is one of the short courses. This paper aims to describe the protocol for evaluating the feasibility and effectiveness of Fit4Study, a behaviour change evidence-informed physical activity intervention for promoting mental health and wellbeing in undergraduate university students.

The primary research question is:

1. What is the effect of Fit4Study on undergraduate students' mental health symptoms (anxiety, depression, wellbeing, and loneliness) when compared to a no-exposure control group, with measurements taken at baseline and at the end of the 4-week intervention?

The secondary research questions are:

2. What is the effect of Fit4Study on undergraduate students' physical activity and sedentary behaviour when compared to a no-exposure control group, with measurements taken at baseline and at the end of the 4-week intervention?
3. What is the effect of Fit4Study on undergraduate students' knowledge and use of physical activity as a coping strategy when compared to a no-exposure control group, with measurements taken at baseline and at the end of the 4-week intervention?
4. What is the feasibility of Fit4Study (reach, retention, and acceptability)?

We hypothesise that participants in the 4-week Fit4Study intervention will report fewer symptoms of anxiety, depression, and loneliness **and** higher psychological wellbeing **(primary outcomes) and an increase in knowledge and use of physical activity as a coping strategy, and higher** physical activity levels, and lower sedentary **behaviour (secondary outcomes), when** compared to the no-intervention control group, as measured at baseline and at the end of the intervention.

2. Methods

2.1. Study overview and design

Ethics for Fit4Study, the physical activity short course, is approved by a large metropolitan university ethics committee (ethics approval number: #2021/HE000259, v.3.01) and first registered with the Australian New Zealand Clinical Trials Registry (ANZCTR; #ACTRN12622001432707) on 8 November 2022. This protocol describes the repeated-measures, non-randomised controlled trial used to evaluate the Fit4Study short course. Fit4Study participants will have their outcome measures compared before and after the intervention when compared to a sample of no-exposure control participants across a similar time frame. Qualitative data to inform feasibility will be collected after the four-week intervention from the Fit4Study participants who opt-in to the focus group sessions.

2.2. Participants

2.2.1. Fit4Study intervention condition

Participants will be a convenience sample of undergraduate students who will be invited to opt in to the course via the Sharper Minds website. The inclusion criteria will require participants to (1) be enrolled at the university where the research is conducted, (2) be at least 18 years old, and (3) possess sufficient English proficiency. Exclusion criteria will include (4) currently receiving treatment for a serious mental health condition and (5) participating in any other Sharper Minds short courses.

2.2.2. Control condition

Control group participants will be drawn from a sample of undergraduate students who opt in for the no-exposure control group as part of the Sharper Minds project. A match-control sequence in SPSS (IBM, 2023) will be conducted based on demographics: age, gender, and enrolment status (international or domestic). Qualitative data will not be collected for control group participants.

2.3. Intervention

Fit4Study is a physical activity-focused short course designed to improve undergraduate students' mental health and wellbeing. Table 1 details the course content and provides examples. Table 2 summarises the behaviour change techniques used in each major component of the intervention, mapped onto the BCTTv1 (Michie et al., 2013). This four-week intervention contains both asynchronous content, provided to participants via a website link before the first session, and synchronous

Table 1
Description and examples of Fit4Study content.

	Example
Week 1 – Introduction, Tracking, and Goal Setting	
Asynchronous	
Definition of physical activity (PA)	
Recommended levels of sufficient PA	150 min/week of moderate intensity PA
Different ways to measure PA intensity	Based on Heart Rate or perceived rate of exhaustion
Benefits (physical and mental) of sufficient PA	Improved cardiovascular health, decreased likelihood of experiences of depression and suicidality
Introduction to Transtheoretical Model of Behavioural Change	(Prochaska, Redding, & Evers, 2002)
How to track PA	Smart watch, pen and paper, etc.
How to breathe when engaging in PA	
Synchronous	
Recap and questions about asynchronous content	
Introduction to SMART Goals	Smart, Measurable, Achievable, Realistic, Timebound
Goal Setting	Based on individual PA goals
Optional bodyweight exercise circuit	Led by facilitator (push ups, sit ups, squats)
Week 2 – Aerobic/Anaerobic Exercise & Stretching	
Asynchronous	
Definition, benefits, and examples of aerobic exercise	Definition: Using oxygen to produce energy Benefit: Improves cardiovascular and respiratory function Example: Running, swimming, and cycling
Definition, benefits, and examples of anaerobic exercise	Definition: Does not use oxygen to produce energy – quick bursts of energy Example: Sprinting, weightlifting, jumping
Explanation and benefits of high-intensity interval (HIIT) exercise	Explanation: 4–20 mins per session, 80 % maximum heart rate. Benefits: Saves time and does not require equipment
Benefits and examples of machine-based cardiovascular exercises	Benefits: helps isolate muscle if an injury exists Examples: Treadmill, bicycle machine, rowing machine.
Benefits and examples of bodyweight cardiovascular exercises	Benefits: free, require fewer resources Examples: high-knees, burpees, mountain climbers
Benefits and examples of Stretching	Benefits: injury prevention, helps mobility and posture Examples: dynamic and static stretching
Synchronous	
Update on progress of SMART goals and recap of asynchronous content	Each participant to share with the group about their SMART goals and ask questions to clarify asynchronous content
Demonstration and practice of aerobic/anaerobic exercises, HIIT circuit, machine-based, and bodyweight cardiovascular exercises by Exercise Physiologist/Physiotherapist	Guided by participants' personal interest and goals
Optional cardiovascular exercise circuit	Dynamic stretching warm up 30 s ON, 10 s OFF, HIIT circuit Static stretching cool down
Week 3 – Strength & Balance	
Asynchronous	
Benefits of strength training	Benefits: build strength, reduce risks of injury
Bodyweight strength training benefits and examples	Benefits: convenient, cost-effective, improves muscle function Examples: squats, push ups, pull ups, dips
Machine-based strength training benefits and examples	Benefits: increase muscle size and strength, can isolate muscle, lower risks of injury Examples: lateral pull down, chest press machine, cable machine
Free weight strength training benefits and examples	Benefits: can target multiple muscles at once, great for strengthening core while doing, improves function for everyday activities Examples: dead lifts, bench press, dumbbell lateral raises
Definition, benefits, and examples of balance benefits and exercises	Benefits: stable joints, improves coordination, helps prevent injuries and falls Examples: Yoga, Tai Chi, Single leg exercises, eyes closed exercises, Bosu ball exercises
Synchronous	
Update on progress of SMART goals and incorporating previous week's content, recap of asynchronous content	Each participant to share with the group about their SMART goals and ask questions to clarify asynchronous content
Demonstration and practice of strength training and balance exercises by Exercise Physiologist/Physiotherapist	Guided by participants' personal interest and goals
Optional strength and balance exercise circuit	
Week 4 – Balanced PA Planning	
Asynchronous	
Introduction to Frequency, Intensity, Time, and Type (FITT) Principles	Frequency – number of times engaged in PA per week; Intensity – light, moderate, vigorous; Type – muscular endurance, hypertrophy, strength, power, balance, flexibility; Time – time of day and duration of each session
Synchronous	
Update on incorporating previous week's content, recap of asynchronous content	
Final update and comments on SMART goals	

content delivered by a facilitator in four one-hour group sessions that cover a variety of topics such as self-monitoring of PA, benefits, and examples of common types of PA, and discussion of behaviour change techniques. The Fit4Study intervention was initially developed by two experts in exercise physiology (EB), physiotherapy and behaviour change (SRG) using evidence-based frameworks, to equip university students with knowledge and skills for physical activity participation. The content was also initially developed in conjunction with two second year honours students in clinical exercise physiology and sport science who ensured the language used in the online material was written at the

Table 2
Fit4Study intervention components, COM-B intervention functions, and behaviour change techniques based on Michie et al. (2013).

Intervention Function	Behaviour Change Technique (Codes)	Component of Fit4Study
Enablement	Goal Setting (behaviour and outcome) (1.1, 1.3)	SMART Goal setting, creating PA-specific SMART goals, which include specific routines.
	Action Planning (1.4)	Selecting a PA goal and planning via SMART goals.
	Problem solving (1.2)	Participants encouraged to brainstorm ideas to overcome barriers of PA and how to incorporate PA into daily living despite busy schedules.
Persuasion	Review of outcome goal(s) (1.7)	Reflection on SMART goal progress throughout the week and in-between four sessions.
	Discrepancy between behaviour and goals (1.6)	Participants encouraged to track and report their PA for the past week and are reminded of their SMART goals. Discussions surrounding how to bridge the gap between their current state and goals are part of weekly sessions.
Education	Feedback on Behaviour + Self-Monitoring of Behaviour (2.2, 2.3)	Participants are taught how to track their PA and are asked to report back on their PA behaviour every week.
Training		Reflection on strategies they used that helped them engage in more PA.
Enablement		Facilitators lead a time of reflection on how participants felt after engaging in PA.
Enablement	Social Support (3)	Intervention is run in a group – Group-based intervention also encourages accountability. Participants were encouraged to meet up with one another outside of the 4 sessions.
Incentivisation		Participants also encouraged to find other friends/interest groups for relevant PA so that they can form social connections while participating in PA.
Education + Enablement	Instruction on how to perform behaviour from a credible source (4.1, 9.1)	Online content provides step by step video guides on common types of PA.
Training + Enablement		Facilitator (with relevant qualification) provides in-person verbal instructions and feedback corrections to participants' form.
Education + Training + Modelling	Demonstration of behaviour + Behavioural practice (6.1, 8.1)	Online content provides demonstration. Facilitator also demonstrates various types of PA (e.g., stretches, balance exercises, how to use strength training machines and perform barbell exercises).
Enablement + Education		Participants get to practice the skills being demonstrated.
Training		A short session of PA is organised based on the relevant PA topic of the week as a chance for participants to immediately practice what they have learnt (e.g., facilitators will run a HIIT session that starts with dynamic stretching and ends with static stretching on week 2, which focuses on aerobic exercises and stretching).
Education + Incentivisation	Information about health consequences (5.1)	Participants are also informed on benefits of engaging in specific types of PA (e.g., aerobic exercise lead improved cardiovascular health; machine-based weight training reduces risks of injury, etc.).
Incentivisation + Persuasion		Benefits (physical and mental health) of consistent PA are brought up again by facilitator and discussed within the group. Specific benefits to university life are brought up (e.g., lower stress, better concentration, good form of break from studying).
Enablement	Restructuring of environment + Adding objects to environment (12.1, 12.2, 12.5)	Participants encouraged to determine their access to PA equipment (e.g., public exercise area on their way home, gymnasium access in their apartment).
Environmental Restructuring		Participants are also given tips on access to affordable PA equipment/PA areas around university (e.g., participants encouraged to adopt active forms of transportation by alighting bus a stop earlier and walking or using the bicycle instead of driving).
Education + Persuasion + Coercion + Incentivisation	Information about antecedents (4.2)	Participants are informed about consequences (physical health related) of high levels of sedentary behaviour (e.g., increased risks of cardiovascular-related health risks).
Education + Persuasion + Coercion + Incentivisation		Participants are informed about consequences (university life related) of high levels of sedentary behaviour (e.g., higher amounts of stress due to long periods of studying without break).

appropriate level of detail for university students. This initial iteration was guided by the COM-B model of behaviour change (Michie et al., 2011) and the associated Behaviour Change Technique Taxonomy (BCTTv1; Michie et al., 2013) and underwent pilot testing with undergraduate students, with continuous improvement made based on student feedback. Mapping of the behaviour change techniques to intervention functions was completed by breaking down each component of the Fit4Study intervention and using Michie et al.'s (2014) guide to designing interventions. For this study, Fit4Study will be delivered by the primary author, who has a background in psychology and is trained in behavioural-change techniques, alongside a registered physiotherapist. Weeks 1 and 4 will be held online while weeks 2 and 3 will be facilitated in-person in a space with access to gym equipment (e.g., public park with exercise equipment or gymnasium).

2.4. Procedure

Participants will be recruited from a large metropolitan university in Australia, predominantly through two first-year psychology courses, flyers, word of mouth, and posters put up around the university and on social media. At recruitment, prospective participants will be given the option to either opt in for Fit4Study, join one of the other health behaviour programs from the Sharper Minds package, or choose to be part of the no-exposure control group. After indicating their preferred option, participants will be contacted by the primary author (KH) to complete written informed consent and pre-intervention measures. Participants will be recruited through an opt-in manner, across the first half of each semester. Course groups are expected to range in size between 4 and 10 participants. Recruitment for course groups will continue until the minimum number for power is achieved. A link to the Fit4Study website will also be sent, which will contain the asynchronous content, along with details of what will be included in each of the four sessions. Neither the participants nor the facilitators will be blinded. Participants enrolled in the first-year psychology courses will be given course credits in exchange for participation. Other participants not enrolled in these courses will be entered into a prize draw to win one of 10 $30 cash vouchers. Quantitative data will be collected using self-report measures at baseline and post-intervention. Qualitative data will be collected using focus groups that will be conducted at the end of the intervention. Fit4Study participants will not be able to re-do the same program but will have the option to join another Sharper Minds program in subsequent semesters. In the event of adverse events, a record will be kept for review.

2.5. Outcomes

Primary outcomes will focus on measures related to mental health, including symptoms of depression, anxiety, psychological well-being, and loneliness. Secondary outcomes, such as self-reported physical activity, sedentary behaviour, and the use of physical activity to support mental health and wellbeing will be collected from both Fit4Study and control group participants. Table 3 lists the primary and secondary outcome measures. Demographic data will be collected at Time 1, including information such as age, gender, living situation, faculty, international student status, and ethnic background. Self-report measures will be collected at both Time 1 and Time 2 from Fit4Study and control group participants via surveys to evaluate the effectiveness of Fit4Study as a physical activity intervention for university students' mental health and well-being, compared to a no-exposure control group. Qualitative outcomes will assess the feasibility of the intervention in terms of reach, retention, and acceptability of Fit4Study and will only be collected from Fit4Study participants at Time 2.

Table 3
Overview of Fit4study intervention measures.

Primary Outcomes	Measure	Number of Items
Anxiety Symptoms	PsyCheck is a 20-item questionnaire measuring psychological distress. Scores 4/4 indicate presence of symptom.	4: PsyCheck Items 4, 5, 6, 12
Depression Symptoms	PsyCheck is a 20-item questionnaire measuring psychological distress. Scores 4/4 indicate presence of symptom.	4; PsyCheck Items 9, 11, 13, 16
Mental Wellbeing	Short Warwick-Edinburgh Mental Wellbeing Scale (SWEMWBS).	7
Loneliness	UCLA 8-item Loneliness Scale.	8
Domain Efficacy	Knowledge and Use of Physical Activity to support mental health.	2
Secondary Outcomes		
Physical Activity Levels	Active Australia Survey.	9
Sedentary Behaviour Levels	International Physical Activity Questionnaire (IPAQ).	4
Sociodemographic Variables	Age, Gender, Faculty in University, International Student Status, Existing Treatment for Mental/ Physical Health, marital status.	6
Qualitative Outcomes		
Post-Intervention Focus Group Questions	Theme of Question	
1. What new information did you learn?	Learning outcomes	
2. How might this new information impact your life as a university student? Have you grown in confidence in engaging with PA? How do you think physical activity might help you cope with the stressors as a university student?	Insight into changes in decisional balance and self-efficacy	
3. In terms of program delivery (e.g., *online content, zoom sessions, in-person session, facilitator delivery, clarity of explanation and demonstrations, etc.) what was done well?*	Program qualitative feedback – further contributions to iterative development	
4. What could use some improvements?	Program qualitative feedback – further contributions to iterative development	

2.5.1. Primary outcome measures

Anxiety and Depression. Outcome measures of anxiety and depression will be assessed using subscales from the PsyCheck measure (Jenner et al., 2013; Lee & Jenner, 2010), which is a 20-item screening measure assessing anxiety and depression symptoms. Each subscale is comprised of four items and requires participants to recall whether they had experienced symptoms in the past 30 days and respond with a yes = 1 or no = 0. A sum score from 0 to 4 is tabulated per subscale. Sub-scales of the PsyCheck measures were found to be valid and reliable based on a psychometric study using two samples (N = 1425) of university students' data (Depression: M = 2.83, SD = 2.66, Cronbach's α = 0.82; Anxiety: M = 1.71, SD = 1.30, Cronbach's α = 0.61; Han et al., 2024). The anxiety subscale includes the following items (items 4, 5, 6, 12): "Are you easily frightened?" "Do your hands shake?" "Do you feel nervous?" "Do you find it difficult to make decisions?". The depression subscale includes the following items (items 9, 11, 14, 16): "Do you feel unhappy?" "Do you find it difficult to enjoy daily activities?" "Are you unable to play a useful part in life?" "Do you feel that you are a worthless person?". Internal reliability analyses will be reported. The full 20-item PsyCheck will be administered as part of the broader Sharper Minds project, but only the 8 items relevant to the scoring of depression and anxiety symptoms will be used in the current study.

Psychological Wellbeing. Psychological wellbeing will be assessed using the Short Warwick Edinburgh Mental Wellbeing Survey (SWEMBS; Stewart-Brown et al., 2009). The SWEMBS is a 7-item scale that includes items such as, "I've been feeling optimistic about the future," "I've been thinking clearly," and "I've been feeling close to other people." Participants will be required to indicate their responses on a five-point Likert scale, (1 = none of the time, and 5 = all of the time). A sum of scores between 7 and 35 will provide an indication of wellbeing where higher scores represent better psychological wellbeing. Published means from Australian university students were $M = 23.2$, SD $= 4.8$ for domestic students and $M = 22.7$, SD $= 4.6$ for international students (Dingle et al., 2024). Internal reliability analyses will be reported.

Loneliness. The eight-item UCLA Loneliness Scale (ULS-8; Hays & DiMatteo, 1987) will be used to assess participant loneliness. Participants will complete a self-report survey on a four-point Likert scale (1 = never, 4 = always), comprised of items such as, "I lack companionship," "I feel left out," and "I am unhappy being so withdrawn." Items three, "I am an outgoing person" and six, "I can find companionship when I want it," will be reverse scored. A total score ranging from 8 to 32 will be tabulated. Hays and DiMatteo (1987) previously reported a mean of $M = 16.3$ from a sample of ($N = 199$) university students that had similar gender (61.8 % female) and age (Mage = 21, Rangeage = 17–48) proportions but a lower proportion of Asian students compared to our sample (3.5 % Chinese, 3.5 % other Asian ethnicities). Internal reliability analyses will be reported.

2.5.2. Secondary outcome measures

Physical Activity. The Active Australia Survey (Australian Institute of Health & Welfare, 2003) will be used to assess physical activity. Data will be collected on the number of times, hours, and minutes of walking, vigorous and moderate PA that participants recall over the past week. The scores will be converted into Metabolic Equivalent of Task (MET) minutes, which is a measure used to estimate the total amount of energy expended during physical activity when compared to rest. A PA score in MET minutes/week will be calculated with this formula: [(walking minutes x 3.33 METs) + (moderate-intensity PA minutes x 3.33 METs) + (vigorous intensity PA minutes x 6.66 METs)] (Babaeer et al., 2022). Internal reliability analyses will also be reported.

Sedentary Behaviour. The two sitting items from the International Physical Activity Questionnaire (IPAQ) will be used to measure sedentary behaviour (Craig et al., 2003). These items will assess the total time (in hours and minutes) spent sitting during weekdays and weekends, which will be recalled by participants over the past week. Overall sedentary behaviour will be calculated using the following formula: [(weekday sitting x5) + (weekend sitting x2)/7] (Babaeer et al., 2022). Internal reliability analyses will also be reported.

Use and knowledge of PA to support mental health. Purpose-written items were developed for the Sharper Minds study to measure students' use of health behaviours such as PA to support their mental health (Dingle et al., 2022). Two items will be included in this study: "I regularly use physical activity to support my mental health and well-being" and "I am aware of the link between physical activity and mental health". Participants will be asked to respond to each of these items on a 5-point scale (1 = strongly disagree, 5 = strongly agree). No internal reliability measures will be run because they are single-item questions.

2.6. Feasibility measures

Feasibility will be assessed using reach, retention, and acceptability. Data for reach will be the demographic data (e.g., age, gender, ethnicity, enrolment status, faculty) collected from completers and non-completers of Fit4Study. Participants will have to attend at least 50 % of the sessions to be considered a completer. Retention rates will be measured using attendance, which will be taken before the commencement of every session across the 4-weeks of the intervention. Drop-outs will be noted, and reasons will be recorded if provided.

Qualitative data will be collected from Fit4Study participants as a measure of acceptability via focus group discussions after each group has completed their 4-week intervention. All Fit4Study participants will be invited to attend the focus groups. Only those who choose to opt-in will be included. The sessions will be conducted via Zoom and primarily led by the lead author KH, along with a co-facilitator. No other parties will be present at the time of recording the focus group discussions. To participate in the focus group, participants will be encouraged to find a quiet space and to engage with each other's responses, initiating discussions without prompts from the researchers. Each focus group session is expected to last between 30 and 45 min, with the number of participants per group ranging from 3 to 10. The questionnaire guide was co-developed by the authorship team to measure the acceptability of Fit4Study. Table 3 includes the interview guide and questions that will be asked.

2.7. Sample size

An *a priori* power analysis was conducted using G*Power (Faul et al., 2007) at an alpha level of $p \langle .05$, with power at $\rangle$ 0.8, and a moderate effect size of (cohen's $d = 0.40$). The effect sizes were based on two existing physical activity interventions (Barği, 2022; McFadden et al., 2017) that found positive effects on mental health symptoms with effect sizes ranging between ($d = 0.30$ to $d = 0.93$). The *a priori* power analysis indicated a minimum of 150 participants to reach sufficient power for the quantitative data analysis, with 75 participants in the intervention and control group respectively. Multiple groups of Fit4Study will be run until the minimum number is reached.

2.8. Data analysis plan

A 2 (time; pre, post) x 2 (intervention; Fit4Study, Control) repeated measures ANOVA is planned to be conducted in SPSS to examine whether the Fit4Study intervention is effective in changing primary and secondary outcome measures. A series of between-groups *t*-tests will assess baseline differences in outcome measures. Qualitative data will be analysed using thematic analysis (Braun & Clarke, 2022) following the reflexive thematic analysis guidelines. A concurrent embedded strategy (Creswell & Creswell, 2017) will be utilised to incorporate both types of data, with more weight given to the quantitative data and the qualitative data embedded and providing more comprehensive insights into the overall feasibility of Fit4Study intervention.

3. Discussion

Evidence has demonstrated that physical activity interventions have the potential to improve university students' mental health (Huang et al., 2024). However, interventions should ideally be designed and implemented underpinned by behavioural change theories for them to be replicable and effective (Michie et al., 2009). Physical activity interventions could play a vital role as an early intervention strategy as university students continue to be susceptible to poor mental health (Australian Bureau of Statistics, 2022; Dingle et al., 2022).

The strength of the Fit4Study intervention lies in its innovative design, specifically tailored to improve university students' mental health. We believe that having designed an intervention informed by the COM-B model of behaviour change (Michie et al., 2011) and its accompanying behaviour change techniques as reported in this paper provides a strong foundation for replicability and potential effectiveness as an intervention. This has been previously demonstrated by physical activity interventions aimed at different outcomes (Heeren et al., 2018;

Husband et al., 2019). Another notable strength is that the intervention was iteratively developed based on feedback from university students who had participated in early iterations of the intervention. Gathering feedback ensured that the intervention was fit for purpose. Utilisation of the co-design process has also been demonstrated to be effective by existing health promotion interventions (Martin et al., 2020; Thabrew et al., 2018; Watkins et al., 2024). A potential limitation could be the self-selection sampling process. Participants who choose to join Fit4Study may already have prior experience with or interest in physical activity, which could reduce the pre-post effects during data analysis. However, the self-selection process might also facilitate more effective behaviour change, as participants will be selecting a behaviour they are genuinely interested in Yardley et al. (2020).

In conclusion, if findings provide evidence that Fit4study is indeed effective, the intervention could be instrumental in providing university students with the necessary behaviour change techniques to adopt and maintain physical activity, which will provide positive improvements to their mental health.

CRediT authorship contribution statement

Kevin Huang: Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Emma M. Beckman:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Norman Ng:** Writing – review & editing, Supervision, Methodology, Investigation. **Genevieve A. Dingle:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Sjaan R. Gomersall:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

Data related to this study can be made available upon request.

References

Australian Bureau of Statistics. (2020-2022). *National Study of Mental Health and Wellbeing.* ABS. https://www.abs.gov.au/statistics/health/mental-health/national-study-mental-health-and-wellbeing/latest-release.

Australian Institute of Health and Welfare. (2003). *The active australia survey: A guide and manual for implementation, analysis and reporting.* Canberra: AIHW.

Babaeer, L., Stylianou, M., & Gomersall, S. R. (2022). Physical activity, sedentary behavior, and educational outcomes among Australian university students: Cross-sectional and longitudinal associations. *Journal of Physical Activity and Health, 19*(3), 211–222. https://doi.org/10.1123/jpah.2021-0535

Barği, G. (2022). Effectiveness of physical activity counseling in university students educated by distance learning during COVID-19 pandemic: A randomized-controlled trial. *Journal of Basic and Clinical Health Sciences, 6*(2), 374–384. https://doi.org/10.30621/jbachs.1027410

Braun, V., & Clarke, V. (2022). *Thematic analysis: A practical guide*. Sage.

Craig, C. L., Marshall, A. L., Sjöström, M., Bauman, A. E., Booth, M. L., Ainsworth, B. E., Pratt, M., Ekelund, U., Yngve, A., Sallis, J. F., & Oja, P. (2003). International physical activity questionnaire: 12-country reliability and validity. *Medicine and science in sports and exercise, 35*(8), 1381–1395. https://doi.org/10.1249/01.MSS.0000078924.61453.FB

Creswell, J. W., & Creswell, J. D. (2017). *Research design: Qualitative, quantitative, and mixed methods approaches.* Sage publications.

Dingle, G. A., Hodges, J., Hides, L., McKimmie, B., Gomersall, S., Beckman, E., Birch, S., Smith, S., Zurynski, Y., Maccallum, F., Bentley, S., Wright, O., Walter, Z., & Alhadad, S. (2021). *Sharper Minds 2021 Pilot Report*. https://doi.org/10.31234/osf.io/vy8pw

Dingle, G. A., Han, R., Huang, K., Alhadad, S. S. J., Beckman, E., Bentley, S. V., Edmed, S., Gomersall, S., Hides, R., Lorimer, L., Maccallum, N., McKimmie, F., M, B., Ng, N., Rossa, K., Smith, S. S., Walter, Z. C., Williams, E., Wright, O. R. L., & Tanksale, R (2024a). *Sharper minds: Feasibility and effectiveness of a mental health promotion package for university students targeting multiple self-care behaviours. [manuscript in review]*. The University of Queensland.

Dingle, G. A., Han, R., Alhadad, S. S., Beckman, E., Bentley, S. V., Gomersall, S. R., Hides, L., Maccallum, F., McMkimmie, B. M., Rossa, K., Smith, S. S., Walter, Z. C., Williams, E., & Wright, O. (2024b). Data from four consecutive cohorts of students in Australia (2019–2022) show the impact of the COVID-19 pandemic on domestic and international university students' mental health. *Australian & New Zealand Journal of Psychiatry, 58*(6), 528–536. https://doi.org/10.1177/00048674241233111

Dingle, G., Hodges, J., Hides, L., McKimmie, B.M., Gomersall, S., Beckman, E., Birch, S., Smith, S., Zurynski, Y., Maccallum, F., Bentley, S., Wright, O., Walter, Z., & Alhadad, S.S.J. (2022). *Sharper Minds 2021 Pilot Report*. https://doi.org/10.31234/osf.io/vy8pw.

Faul, F., Erdfelder, E., Lang, A. G., & Buchner, A. (2007). G* Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior research methods, 39*(2), 175–191.

Han, R., Walter, Z. C., Maccallum, F., & Dingle, G. A. (2024). *Evaluating the psychometric properties of the psycheck screening tool for mental health concerns in university students [unpublished manuscript]*. The University of Queensland.

Hays, R. D., & DiMatteo, M. R. (1987). A short-form measure of loneliness. *Journal of Personality Assessment, 51*(1), 69–81.

Heeren, G. A., Jemmott, J. B., Marange, C. S., Rumosa Gwaze, A., Batidzirai, J. M., Ngwane, Z., Mandeya, A., & Tyler, J. C. (2018). Health-promotion intervention increases self-reported physical activity in sub-saharan african university students: A randomized controlled pilot study. *Behavioral Medicine, 44*(4), 297–305. https://doi.org/10.1080/08964289.2017.1350134

Hrafnkelsdottir, S. M., Brychta, R. J., Rognvaldsdottir, V., Gestsdottir, S., Chen, K. Y., Johannsson, E., Guðmundsdottir, S. L., & Arngrimsson, S. A. (2018). Less screen time and more frequent vigorous physical activity is associated with lower risk of reporting negative mental health symptoms among Icelandic adolescents. *PLOS ONE, 13*(4), Article e0196286. https://doi.org/10.1371/journal.pone.0196286

Huang, K., Beckman, E. M., Ng, N., Dingle, G. A., Han, R., James, K., Winkler, E., Stylianou, M., & Gomersall, S. R. (2024). Effectiveness of physical activity interventions on undergraduate students' mental health: Systematic review and meta-analysis. *Health Promotion International, 39*(3), daae054. https://doi.org/10.1093/heapro/daae054

Husband, C. J., Wharf-Higgins, J., & Rhodes, R. E. (2019). A feasibility randomized trial of an identity-based physical activity intervention among university students. *Health Psychology and Behavioral Medicine, 7*(1), 128–146. https://doi.org/10.1080/21642850.2019.1600407

IBM Corp. (2023). *IBM spss statistics for windows (Version 29.0) [Computer software]*. IBM Corp.

N. Jenner, L., Cameron, J., Lee, K., & Nielsen, S (2013). Test-retest reliability of PsyCheck: A mental health screening tool for substance use treatment clients *Advances in Dual Diagnosis, 6*(4), 168–175

Lee, N. K., & Jenner, L. (2010). Development of the PsyCheck screening tool: An instrument for detecting common mental health conditions among substance use treatment clients. *Mental Health and Substance Use: Dual Diagnosis, 3*(1), 56–65.

Martin, A., Caon, M., Adorni, F., Andreoni, G., Ascolese, A., Atkinson, S., Bul, K., Carrion, C., Castell, C., Ciociola, V., Condon, L., Espallargues, M., Hanley, J., Jesuthasan, N., Lafortuna, C. L., Lang, A., Prinelli, F., Puidomenech Puig, E., Tabozzi, S. A., & McKinstry, B. (2020). A mobile phone intervention to improve obesity-related health behaviors of adolescents across europe: Iterative co-design and feasibility study. *JMIR mHealth and uHealth, 8*(3), e14118. https://doi.org/10.2196/14118

McFadden, T., Fortier, M. S., & Guérin, E. (2017). Investigating the effects of Physical Activity Counselling on depressive symptoms and physical activity in female undergraduate students with depression: A multiple baseline single-subject design. *Mental Health and Physical Activity, 12*, 25–36. https://doi.org/10.1016/j.mhpa.2017.01.002

Michie, S., Fixsen, D., Grimshaw, J. M., & Eccles, M. P. (2009). Specifying and reporting complex behaviour change interventions: The need for a scientific method. *Implementation Science, 4*(1), 40. https://doi.org/10.1186/1748-5908-4-40, 1748-5908-4–40.

Michie, S., Van Stralen, M. M., & West, R. (2011). The behaviour change wheel: A new method for characterising and designing behaviour change interventions. *Implementation Science, 6*(1), 42. https://doi.org/10.1186/1748-5908-6-42

Michie, S., Richardson, M., Johnston, M., Abraham, C., Francis, J., Hardeman, W., & Wood, C. E. (2013). The behavior change technique taxonomy (v1) of 93 hierarchically clustered techniques: Building an international consensus for the reporting of behavior change interventions. *Annals of behavioral medicine, 46*(1), 81–95.

Michie, S., Atkins, L., & West, R. (2014). *The behaviour change wheel: A guide to designing interventions.* Silverback.

Prochaska, J. O., Redding, C. A., & Evers, K. (2002). The Transtheoretical Model and Stages of Change. In K. Glanz, B. K. Rimer, & F. M. Lewis (Eds.), *Health Behavior and Health Education: Theory, Research, and Practice* (3rd Ed.). San Francisco, CA: Jossey-Bass, Inc.

Reyes-Molina, D., Alonso-Cabrera, J., Nazar, G., Parra-Rizo, M. A., Zapata-Lamana, R., Sanhueza-Campos, C., & Cigarroa, I. (2022). Association between the physical activity behavioral profile and sedentary time with subjective well-being and mental health in chilean university students during the COVID-19 pandemic. *International Journal of Environmental Research and Public Health, 19*(4), 2107. https://doi.org/10.3390/ijerph19042107

Román-Mata, S. S., Puertas-Molero, P., Ubago-Jiménez, J. L., & González-Valero, G. (2020). Benefits of physical activity and its associations with resilience, emotional intelligence, and psychological distress in university students from southern spain. *International Journal of Environmental Research and Public Health, 17*(12), 4474. https://doi.org/10.3390/ijerph17124474

Stewart-Brown, S., Tennant, A., Tennant, R., Platt, S., Parkinson, J., & Weich, S. (2009). Internal construct validity of the Warwick-Edinburgh mental well-being scale (WEMWBS): A Rasch analysis using data from the Scottish health education population survey. *Health and quality of life outcomes, 7*, 1–8.

Thabrew, H., Fleming, T., Hetrick, S., & Merry, S. (2018). Co-design of eHealth interventions with children and young people. *Frontiers in Psychiatry, 9*, 481. https://doi.org/10.3389/fpsyt.2018.00481

Watkins, J. M., Brunnemer, J. E., Heeter, K. N., Medellin, A. M., Churchill, W. C., Goss, J. M., Hobson, J. M., Werner, N. E., Weaver, R. G., Kercher, V. M. M., & Kercher, K. A. (2024). Evaluating the feasibility and acceptability of a co-designed physical activity intervention for rural middle schoolers: A pilot study. *BMC Public Health, 24*(1), 1830. https://doi.org/10.1186/s12889-024-19356-2

Yardley, L., Morrison, L., Muller, I., & Bradbury, K. (2020). Methods and processes of behaviour change: Intervention development, application, and translation. In M. S. Hagger, L. D. Cameron, K. Hamilton, N. Hankonen, & T. Lintunen (Eds.), *The handbook of behaviour change* (pp. 267–442). Cambridge University Press. https://doi.org/10.1017/9781108677318.